Software Engineering I
Course Policies

Lecture 1a

Agenda

- About the Lecturer
- About the Course
- Marking Scheme
- Key Dates
- Resources for Course
- Project Considerations
- General Guidelines

About the Lecturer

Personal Information

Carlos Mera Gómez

- Academic details
  - PhD in Computer Science from The University of Birmingham, UK.
  - MSc. Internet Software Systems from The University of Birmingham, UK.
  - Computer Engineer, ESPOL, Ecuador.
- Professional Experience
  - Operations Director at ECU-911.
  - Technological Advisor at Ministry of Mines and Oil.
  - Technological Consultant at Ministry of Defense.
  - Software Architect at Ecuadorian Maritime Authority.
  - Software Developer at Ecuadorian Navy.

Personal Information

Carlos Mera Gómez

- Academic Experience
  - Lecturer at ESPOL since 2012
    - ❖Software Engineering, Software Design, Computer Networks, Programming Fundamentals at undergraduate level.
    - ❖Advanced Topics of Software Engineering at postgraduate level.
  - Honorary Research Fellow at The University of Birmingham (2023 - 2029)
  - Teaching Assistant at ESPOL for 3 years: Statistics, Physics, Maths.

Interests

- Research
  - Self-Adaptive and Autonomous Distributed Systems
  - Technical Debt
  - Economics-Driven Software Engineering
  - Cloud Elasticity Management
- Personal
  - Geopolitics
  - Corporate and Transnational Corruption
  - Astrophysics
  - Medicine, Health and Nutrition
  - Genetic Engineering

Contact Information

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About the Course

Course Description

- This course addresses the importance of software engineering, the development life-cycle of a software product, and the application of a software development methodology.
  - Specifically, it makes emphasis on planning, management, requirements analysis and design of a software system of medium complexity.
- Technical and non-technical skills are developed or improved during the course.
  - Among the technical skills we have: the application of methods and tools to plan a software development project, the analysis of the needs of a real customer, and the design of a software system: both architectural and detailed design.
  - The non-technical skills include teamwork; ethics in the software engineering profession; and oral and written communication.

Course Goals

- To identify a software development process model that adds value to a specific project by analysing the characteristics of existing models and the specificities of the project.
- To organize the development of a software system through the creation of work teams considering roles and responsibilities.
- To apply standards and best practices in requirements engineering for the generation of software requirements specifications for a system of medium complexity.
- To create design models, based on software requirements specifications, that represent the structure and behaviour of a software system.

Student Outcomes

2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

Course Content

- For the First Evaluation:

1. Procesos de software
1.1. La programación individual vs. la programación de grandes sistemas.
1.2. Ingeniería de software: ¿Qué es? ¿Cuál es su importancia?
1.3. El ciclo de vida de desarrollo de software.
1.4. Introducción a los modelos de proceso de software (cascada, incremental, espiral, iterativos, ágiles).
1.5. Evaluación de modelos de proceso de software.

4. Ingeniería de requerimientos
4.1. Conceptos básicos de ingeniería de requerimientos de software: propiedades e importancia de requerimientos de software, tanto funcionales como no funcionales.
4.2. Técnicas para el levantamiento y análisis de requerimientos de software.
4.3. Descripción y documentación de requerimientos de software.
4.4. Prototipado de requerimientos funcionales.
4.5. Verificación de especificaciones de requerimientos funcionales de software.

2. Administración de proyectos de software
2.1. Roles y responsabilidades en un equipo de desarrollo de software.
2.2. Gestión del equipo de desarrollo de software: organización, asignación de recursos y responsabilidades.
2.3. Medición de software y estimación de esfuerzo.

Course Content

- For the Second Evaluation:

2. Administración de proyectos de software
2.4. Análisis de riesgos.
2.5. Gestión del proyecto de software: planificación de actividades, monitoreo.
3. Herramientas y ambientes de desarrollo de software
3.1. Herramientas para la planificación y monitoreo de proyectos de desarrollo de software.
3.2. Herramientas para el modelamiento de prototipos o "mockups" de requerimientos funcionales de software.
3.3. Herramientas para el diseño de software.
5. Diseño de software
5.1. Revisión de principios y criterios de diseño de software.
5.2. Conceptos de arquitectura de software y estilos arquitectónicos (cliente-servidor, capas, tubos y filtros, etc.).
5.3. Modelos de diseño de software estructurales y de comportamiento usando el Lenguaje de Modelamiento Unificado (UML).
6. Ética en la profesión
6.1. El impacto de mal software en la sociedad.
6.2. Los principios éticos de la ingeniería de software (códigos de ética).
6.3. Dilemas éticos: argumentación ética, análisis de responsabilidades y consecuencias.

Marking Scheme

Marking Scheme

- The weighted sum of the marks is as follows:

Activity		First Evaluation	Second Evaluation	Third Evaluation	
EHD/Theoretical	Examination	35% Ω	35%	35%	70%**
EHTA/Autonomous	Project δ	30% *	30% *	35% *	--
	Tests	05%	05%	--	--
EHP/Practical	Workshops	30%		--	

- Ω Through the first half of II PAO 2026
- * An acceptance form signed by the client is compulsory.
- ** If student passed with the first and second evaluation.
- δ The project mark requires a viva, which grants an individual factor between 0 and 1 to every student.

EHD: Evaluación Horas de Docencia
EHTA: Evaluación Horas de Trabajo Autónomo
EHP: Evaluación Horas Prácticas

Key Dates

Timetable

- Sessions are scheduled on Mondays and Wednesdays from 13:00 to 15:00.
- Workshops are scheduled on Wednesdays.

Día	Hora	Aula	Bloque
Mondays	13:00 to 15:00	L204	11A Campus Gustavo Galindo
Wednesdays	13:00 to 15:00	L204	11A Campus Gustavo Galindo

- Office hours:
  - Mondays and Wednesdays from 12:30 to 13:00.
  - Open doors policy.

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2026

Timetable (Course 3)

- Sessions are scheduled on Tuesdays and Thursdays from 13:00 to 15:00.
- Workshops are scheduled on Thursdays.

Día	Hora	Aula	Bloque
Tuesdays	13:00 to 15:00	L204	11A Campus Gustavo Galindo
Thursdays	13:00 to 15:00	L204	11A Campus Gustavo Galindo

- Office hours:
  - Mondays and Wednesdays from 12:30 to 13:00.
  - Open doors policy.

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2026

Key Dates

- Evaluations:
  - The 4th and 7th week; First Evaluation Ω
  - 11 January 2027; Second Evaluation *
  - 25 January 2027; Third Evaluation *

  *Evaluations are scheduled on Mondays from 9:00 to 11:00 and, tentatively, at room A102, block 11A.

  Ω During scheduled weekly sessions.

Key Dates

- Project Deliveries:
  - 07 October 2026 at the latest, Project Approved
  - 04 November 2026, Midterm Project Delivery
  - 04 January 2027, Final Project Delivery
  - 25 January 2027, Extended Project Delivery

Resources for the Course

Bibliography

- Sommerville, Ian. (2015). Software Engineering, Global Edition. (Tenth Edition).
- Pressman Roger, Maxim Bruce. (2019). Software Engineering: A Practitioner's Approach. (Ninth Edition).
- Cohn, Mike (2014). User Stories Applied for Agile Software Development.
- International Scrum Institute. (2019). The Scrum Framework. (Third Edition).

Bibliography

- Rubin, Kenneth S. (2013). Essential Scrum. (First Edition)
- Bass, Clements, and Kazman. (2021). Software Architecture in Practice. (Fourth Edition)
- Taylor, Medvidovic, and Dashofy (2009). Software Architecture: Foundations, Theory, and Practice. (First Edition)

Project Considerations

Project Considerations

- An application software automating business processes must be developed for a real customer. This means…
  - No games
  - No simulators
  - No embedded systems
  - No utility software.
- The customer (company) cannot be (belong to) a member of your family.
- The software must have a Web and mobile components or only a mobile one.

Project Considerations

- Teams composed of up to 5 members.
- Fill the form of project acceptance.
- The project must be completed in two academic terms (Software Engineering I and II).
- The customer of your project must be committed to attend to scheduled meetings with the team members.
- Co-evaluation between team members.

Project Considerations

- The learning of development tools are part of a self-learning process of the student that depends on the specific needs of their customer.
- You are expected to live a real experience and therefore you should satisfy the requirements of your customer.
- The project should be developed in the influence area of Guayaquil.
- The company must have the infrastructure or resources required to put in operation the software system (Software Engineering II).

Co-Evaluation of Teamwork

- Individual grades on group projects will be determined based on a rubric.
  - For example, consider an individual assignment that result from group work or a group assignment (e.g., poster presentation) that earns a grade of 85%. A student who achieves a Level 5 rating will receive the full points (i.e., 85%) earned for the assignment. A student who achieves a Level 2 rating will receive 25% of the points earned for the assignment (i.e., 21.25%).

Co-Evaluation of Teamwork

- **Level 5: 100% of the assignment grade**
  - The team member participated fully in all discussions, contributing your own ideas and suggestions in each phase of the project.
  - The team member participated fully in project activities (planning, data or information collection, data analysis, writing, presenting, etc.).
  - The team member attended all group meetings.
  - The team member completed all tasks that you agreed to conduct for the project/team.

Co-Evaluation of Teamwork

- **Level 4: 75% of the assignment grade**
  - The team member participated in each phase of the project but showed less initiative in contributing your ideas and suggestions.
  - The team member usually participated in project activities, but not always.
  - The team member usually attended team meetings.
  - The team member usually completed tasks that you agreed to conduct for the project/team.

Co-Evaluation of Teamwork

- **Level 3: 50% of the assignment grade**
  - The team member occasionally participated in the various phases of the project, generally as a “follower” rather than contributing to project development
  - The team member occasionally attended team meetings
  - The team member completed one or two minor tasks that you agreed to conduct for the project/team

Co-Evaluation of Teamwork

- **Level 2: 25% of the assignment grade**
  - The team member contributed very little to the project, showing no initiative
  - The team member missed the vast majority of the team meetings
  - The team member failed to complete the majority of tasks that you agreed to conduct for the project/team
- **Level 1: No credit**
  - The team member did not participate in developing the project/presentation

Project Reviews

- Reviews per sprint.
- Each completed sprint must include:
  - An acceptance form signed by the client.
  - A record of each sprint review.

General Guidelines

Attendance

- According to regulations, the attendance is compulsory and must be controlled by the teacher.
- When the number of absences exceeds the forty percent (40%) of the scheduled sessions, the student may fail the module.
- Continuous class participation will be rewarded.
- The mobile must remain put away during class. It cannot be placed on the desk.
- Chats are not allowed once the lecture begins.
- Punctuality is encouraged.

General Guidelines

- Dishonesty (including plagiarism) produces a mark of zero.
- Self-registration in teams for workshops and the project.
- Assessments must be taken on established or agreed dates and cannot be postponed.
- Assignments and workshops will only be received through the Aula Virtual academic management platform.
- Evaluations cannot be recovered under any circumstances

Policy on Authorship (Based on ACM)

- Can I use generative AI software tools to prepare my assignments?
- Yes, the use of such tools is permitted, subject to the following requirements:
  - That these systems do not plagiarize, misrepresent, or falsify content in assignment submissions.
  - That the resulting work in its totality is an accurate representation of the students' underlying work and intellectual contributions and is not primarily the result of the tool's generative capabilities.
  - That the students accept responsibility for the veracity and correctness of all material in their work, including any computer-generated material.

Policy on Authorship (Based on ACM)

- If I use generative AI software tools, under what conditions must I disclose their use?
  - If you are using generative AI software tools such as ChatGPT, Jasper, AI-Writer, Lex, or other similar tools to generate new content such as text, images, tables, code, etc. you must disclose their use.
  - If entire sections of a work, including tables, graphs, images, and other content were generated by one of these tools, you should disclose which sections and which tools and tool versions you used to generate those sections by preparing an Appendix or a Supplementary Material document that describes the use, including but not limited to the specific tools and versions, the text of the prompts provided as input, and any post-generation editing (such as rephrasing the generated text). Students should also note that the amount or type of generated text allowable may vary depending on the type of the section affected.

Policy on Authorship (Based on ACM)

- If I use generative AI software tools, under what conditions must I disclose their use?
  - If you are using generative AI software tools to edit and improve the quality of your existing text in much the same way you would use a typing assistant like Grammarly to improve spelling, grammar, punctuation, clarity, engagement or to use a basic word processing system to correct spelling or grammar, it is not necessary to disclose such usage of these tools in your Work.

Courtesy: “ACM Policy on Authorship”, https://www.acm.org/publications/policies/new-acm-policy-on-authorship

How to Appropriately Write an Email?

- http://es.wikihow.com/escribir-un-e-mail-formal
- http://www.ecoescritura.com/%C2%BFcomo-escribir-correos-electronicos/
- http://www.ehowenespanol.com/reglas-saludo-correo-electronico-lista_43535/

Inquiries

- Inquiries to the Lecturer should be sent either through email or Aula Virtual.
- Additionally, communications should follow the suggested guidelines and keep an excellent orthography.

Next On

Next Lecture

- Team Building
- Software Processes